Number-theoretic algorithms exercises

31-E01 Euclid's certificate and input length

Find $\gcd(391, 299)$, listing quotients, remainders, and Bezout back substitution. Find $299^{-1} \pmod{391}$, or prove it does not exist. Prove that every two Euclid divisions at least halve the second argument. Explain why $O(\log n)$ divisions do not imply $O(\log n)$ bit time.

31-E02 All modular linear solutions

Solve $24x \equiv 36 \pmod{60}$ and $24x \equiv 35 \pmod{60}$. Give complete solution sets and spacings, proving completeness. Solve $0x \equiv 0 \pmod{9}$, then compare compressed output with enumeration.

31-E03 CRT construction and compatibility

Use selectors to solve $x \equiv 1 \pmod{4}, x \equiv 2 \pmod{9}, x \equiv 3 \pmod{5}$. List M_i, inverses, and reconstruction. Solve $x \equiv 3 \pmod{8}, x \equiv 7 \pmod{12}$, then replace 7 with 6. Prove compatibility and the correct period.

31-E04 Orders, primitive roots, and different groups

List every power of 2 modulo 11 and its order. Give $\mathrm{ind}_{11,2}(7)$. Prove $U(8)$ is not cyclic even though it has four elements and every element has an inverse. Construct a primitive root modulo 121 starting from $g = 2 \pmod{11}$. Compute $2^{10} \pmod{121}$ and justify its order. Being a generator modulo 11 alone is insufficient.

31-E05 Left-to-right and right-to-left powers

Trace $7^{19} \pmod{23}$ through all left-to-right bits 10011. Write right-to-left square and multiply and prove its invariant $ra^b \equiv A^B \pmod{n}$ through every update. State behavior for $b = 0, n = 1$, and noncoprime a, n.

31-E06 Nonunit RSA and malleability

For $p = 7, q = 13, e = 5$, compute n, φ, d. Encrypt and decrypt $m = 7$, listing intermediate powers. Use the modulo-7 and modulo-13 components and CRT to prove correctness for this nonunit. For another ciphertext $c = E(m)$, describe its message after multiplication by $2^e \pmod{n}$. State which security conclusions this algebra does not prove.

Number-theoretic algorithms exercises (continued)

31-E07 Count every strong liar

For $n = 65$, find $n - 1 = 2^s d$, all t_i, u_i, D_i, τ, and strong-liar count L. Use CRT to list every passing base in $[1, 64]$. Compute the passing fraction in the actual domain $[2, 63]$. Give the base-2 chain and identify a witness or liar.

31-E08 Random testing and posterior probability

How many independent uniform Miller-Rabin rounds ensure acceptance probability at most 10^{-9} for every fixed composite? If the prior prime probability is $\pi = 1/100$, bound $\Pr(\text{composite} \mid \text{pass})$ for that round count. Does the guarantee hold if all rounds use fixed base 2? Give a composite counterexample.

31-E09 Pollard rho component collisions

For $n = 91, x_0 = 2, f = x^2 + 1$, perform one Floyd round and compute the gcd. For $n = 15, x_0 = 0, f = x^2 + 2$, continue until $d > 1$. Explain why $d = n$ cannot be returned as a successful factor. Prove Floyd's meeting bound for finite functional graphs. State the extra assumption for an $O(\sqrt{p})$ estimate.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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